

Y-TRACE: YOUTH ORGANIZATION REGISTRATION, COMPLIANCE, AND BUDGET MONITORING

SYSTEM FOR LOCAL YOUTH DEVELOPMENT OFFICES

CHAPTER 3: DESIGN SPECIFICATIONS — DATABASE SCHEMA SPECIFICATIONS

Process-Based Entity Relationship Diagrams (ERDs), Relational Architecture, and Complete Coverage Audit

Authoritative Source: Supabase Audit (Sep 15, 2026)	Target Platform: PostgreSQL 15 / Supabase Cloud	Application Tables: 42 of 42 Documented (100%)	Foreign Key Constraints: 65 of 65 Documented (100%)
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1. Database Architecture & Schema Decomposition Rationale

The database architecture of Y-TRACE is built on PostgreSQL 15, hosted and managed via the Supabase cloud platform. To support the multifaceted administrative and regulatory mandates of the Pasig City Local Youth Development Office (PCYDO) — pursuant to the Pasig City Youth Development Code (Ordinance No. 50, Series of 2024) and the Sangguniang Kabataan Reform Act (Republic Act No. 10742 as amended by Republic Act No. 11768) — the database implements a rigorous relational structure comprising forty-two (42) public application tables, 476 typed columns, sixty-five (65) explicit foreign key constraints, twenty-three (23) unique constraints, sixty (60) performance indexes, thirty (30) database automation triggers, 120 stored functions/RPCs, fifty-eight (58) Row-Level Security (RLS) policies, and nine (9) dedicated Supabase Storage buckets.

In software engineering and database design specifications for enterprise and municipal management information systems, presenting forty-two tables within a single monolithic Entity Relationship Diagram creates severe visual overcrowding, connector routing collisions, and diminished readability. Such unsegmented diagrams obscure the distinct operational lifecycles of the application and impede clear relational analysis. Consequently, the Y-TRACE database schema is systematically partitioned into eight (8) cohesive, process-oriented Entity Relationship Diagrams (Figures 8.1 through 8.8). Each diagram directly corresponds to a core business process identified in the system's Data Flow Diagrams (DFDs) and functional requirements specification, while preserving complete relational integrity across the entire schema.

The eight process-based ERD figures encompass every single application table without exception. Standby, reference, and legacy entities — such as organizations, events, programs, and transparency_posts — are transparently integrated and identified with appropriate structural notations, ensuring that the documentation reflects 100% of the active database without silent omission. Tables that participate across multiple distinct workflows (most notably organization_profiles, auth.users, and admin_accounts) are modeled as primary focal entities in their primary operational module and represented as formal [Reference Entities] in dependent diagrams, maintaining consistent table definitions and precise cardinality without redundancy.

1.1. Diagrammatic Conventions and Relational Legend

The Entity Relationship Diagrams conform to the visual style established in the capstone manuscript, featuring white canvas backgrounds, clean orthogonal connectors, and distinct header banners distinguishing functional roles. Data types reflect the audited PostgreSQL types (e.g., uuid, citext, timestamptz, jsonb, numeric(12,2)). The architectural conventions and relationship symbols are summarized in Table 8.0.

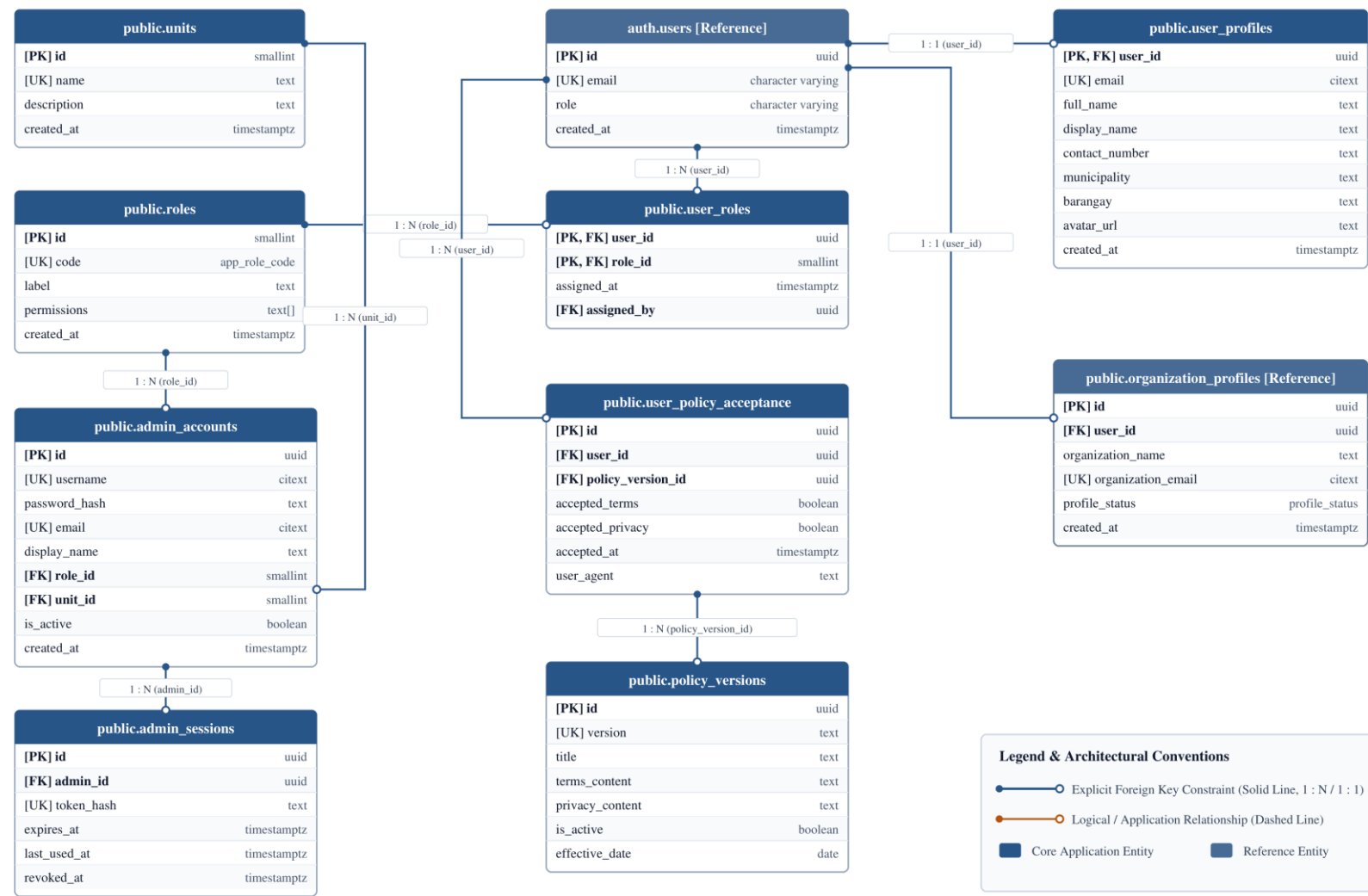
Visual Element	Classification	Architectural & Relational Meaning
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Visual Element	Classification	Architectural & Relational Meaning
Solid Blue Box (#285587)	Core Application Entity	Active production application table created in public schema, owning data records and direct process workflow fields.
Slate Blue Box (#476b94)	Shared / Reference Entity	External or cross-process table (e.g., auth.users, organization_profiles) providing parental foreign key references to child entities.
Brown Box (#8c524b)	Schema Standby / Legacy Entity	Schema table maintained for structural compatibility, future planned modules, or legacy fallback (e.g., organizations, events, move_applications).
Solid Line + Circle Connector	Explicit Database Foreign Key	Physical PostgreSQL FOREIGN KEY constraint enforced by the database engine (Referenced PK has filled circle, Child FK has hollow circle).

Table 8.0. Database Schema Entity and Connector Legend

2.1. Process 7: Manage Administration, Monitoring, and Reporting (Identity & Policy Subsystem)

Figure 8.1. Entity Relationship Diagram of Authentication, Users, Roles, Policies, and Administrative Access



Process & Relational Workflow Narrative: Figure 8.1

Figure 8.1 delineates the dual-tier identity, authentication, role-based access control (RBAC), and policy governance architecture of Y-TRACE. To safeguard municipal operations and protect sensitive organizational records, the system implements a strict separation of user tiers. Youth organization representatives authenticate through the Supabase Authentication service (represented as the external parent entity `auth.users`), which maintains core email/password credentials and session states. Application-level user profile attributes (such as full name, contact number, municipality, and assigned barangay) are isolated in `public.user_profiles`, establishing a 1:1 foreign key binding to `auth.users(id)`. User permissions and platform authorization are managed through `public.roles` and the associative table `public.user_roles`, linking authenticated identities to standardized role codes (`admin`, `staff`, `youth_org`, `guest`).

Internal PCYDO administrative personnel access the administrative portal through an independent, secured credential repository, `public.admin_accounts`. This table maintains dedicated administrative usernames, Bcrypt password hashes, active flags, and explicit foreign key relationships to `public.units` (classifying administrative personnel into PCYDO functional divisions: Administration & Finance, Programs & Operations, Policy & Planning) and `public.roles`. Active administrative sessions are tracked in `public.admin_sessions` using cryptographically hashed tokens, hard expiration timestamps, and revocation tracking to ensure strict session invalidation upon sign-out.

Regulatory compliance and data privacy adherence under Republic Act No. 10173 are enforced via `public.policy_versions` and `public.user_policy_acceptance`. Before accessing protected portal features, authenticated organization representatives must accept the active Terms of Service and Privacy Policy. Each acceptance event is immutably recorded with policy version keys, IP hash, user agent string, and acceptance timestamps, creating an auditable legal compliance trail.

Table Inventory for Figure 8.1

Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
units	Smallint lookup table defining PCYDO administrative units/divisions.	id (smallint)	None	N/A	3
roles	System role master table defining authorization tiers and permissions array.	id (smallint)	None	N/A	4
admin_accounts	Dedicated credentials and profiles for internal PCYDO staff and administrators.	id (uuid)	role_id -> roles.id, unit_id -> units.id	N/A	5
admin_sessions	Server-side administrator session tokens, validity windows, and revocation flags.	id (uuid)	admin_id -> admin_accounts.id	N/A	241
user_profiles	Extended biographical and contact profile metadata for authenticated users.	user_id (uuid)	user_id -> auth.users.id	avatars	20
user_roles	Junction table mapping authenticated user identities to application roles.	(user_id, role_id)	user_id -> auth.users.id, role_id -> roles.id, assigned_by -> auth.users.id	N/A	20
policy_versions	Terms of Service and Privacy Policy markdown content and version history.	id (uuid)	None	N/A	3
user_policy_acceptance	User acceptance records for active legal terms and privacy notices.	id (uuid)	user_id -> auth.users.id, policy_version_id -> policy_versions.id	N/A	9

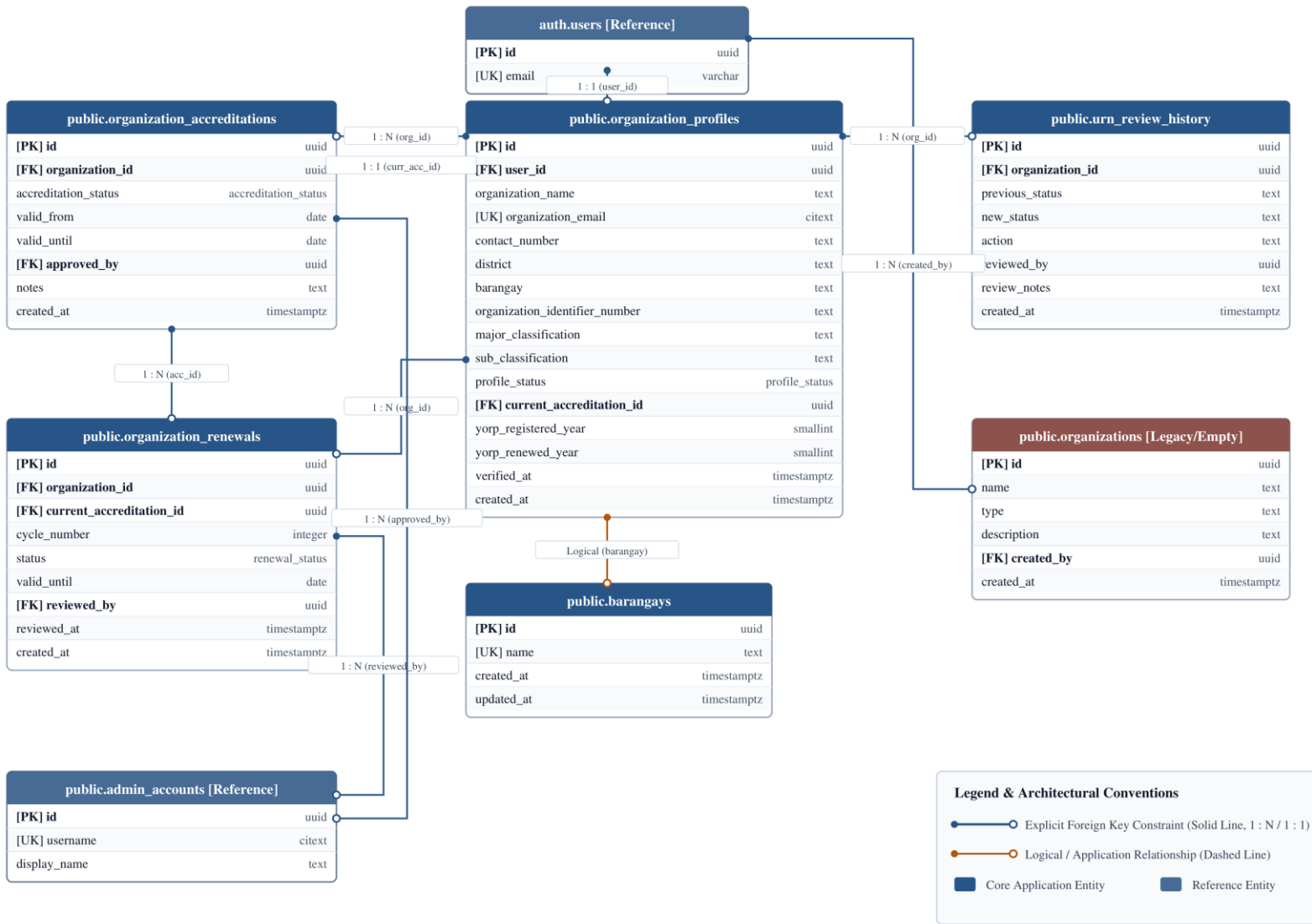
Relationship Catalog for Figure 8.1

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
admin_accounts.role_id	roles.id	admin_accounts_role_id_fkey	Explicit FK	Many-to-One	NO ACTION / NO ACTION
admin_accounts.unit_id	units.id	admin_accounts_unit_id_fkey	Explicit FK	Many-to-One	NO ACTION / NO ACTION

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
admin_sessions.admin_id	admin_accounts.id	admin_sessions_admin_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
user_profiles.user_id	auth.users.id	user_profiles_user_id_fkey	Explicit FK	One-to-One	CASCADE / NO ACTION
user_roles.user_id	auth.users.id	user_roles_user_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
user_roles.role_id	roles.id	user_roles_role_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
user_roles.assigned_by	auth.users.id	user_roles_assigned_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
user_policy_acceptance.user_id	auth.users.id	user_policy_acceptance_user_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
user_policy_acceptance.policy_version_id	policy_versions.id	user_policy_acceptance_policy_version_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

2.2. Process 1: Manage Organization Registration and Renewal

Figure 8.2. Entity Relationship Diagram of Organization Registration, Renewal, Accreditation, and URN Management



Process & Relational Workflow Narrative: Figure 8.2

Figure 8.2 illustrates the core organizational lifecycle subsystem governing registration, verification, official accreditation, and Unique Registration Number (URN) administration. The foundational anchor of this subsystem is `public.organization_profiles`, which maintains the master municipal registry of all registered Youth Organizations (YOs) and Youth-Serving Organizations (YSOs) in Pasig City. Each profile is established by an authenticated user (`auth.users.id`) and records institutional classification (major classification, sub-classification, advocacies array), contact parameters, representative details, and geographical location within one of Pasig City's thirty barangays (referenced via `public.barangays`).

Official municipal recognition is formalized through `public.organization_accreditations`. When PCYDO administrators approve an organization's qualifying credentials, an accreditation record is created containing the assigned accreditation number, validity dates, certificate issuance metadata, and the foreign key of the approving administrator (`admin_accounts.id`). The active accreditation is mirrored in `organization_profiles` via `current_accreditation_id`. To monitor multi-year institutional continuity, `public.organization_renewals` captures periodic renewal applications, document inspection outcomes, and administrative review decisions.

The issuance and lifecycle of Unique Registration Numbers (URNs) — mandated under National Youth Commission (NYC) YORP guidelines — are tracked in `public.urn_review_history`, documenting the transition of URN states (pending, approved, flagged, expired) and retaining administrative remarks. Finally, `public.organizations` is maintained in the schema as a legacy standby table to support backwards compatibility with preliminary data migrations.

Table Inventory for Figure 8.2

Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
organization_profiles	Master profile and registry record for YORP-accredited youth organizations.	id (uuid)	user_id -> auth.users.id, current_accreditation_id -> organization_accreditations.id	N/A	6
organization_accreditations	Formal accreditation certificates, validity windows, and approval metadata.	id (uuid)	organization_id -> organization_profiles.id, approved_by -> admin_accounts.id	N/A	5
organization_renewals	Annual YORP renewal application records, review stages, and approval trails.	id (uuid)	organization_id -> organization_profiles.id, current_accreditation_id -> organization_accreditations.id, reviewed_by -> admin_accounts.id	N/A	0
urn_review_history	Historical audit trail of URN evaluations, status changes, and administrator remarks.	id (uuid)	organization_id -> organization_profiles.id	N/A	0
barangays	Master territorial registry of Pasig City barangays, coordinates, and demographic metrics.	id (uuid)	None	N/A	3
organizations	Legacy organization repository retained for database migration compatibility.	id (uuid)	created_by -> auth.users.id	N/A	0

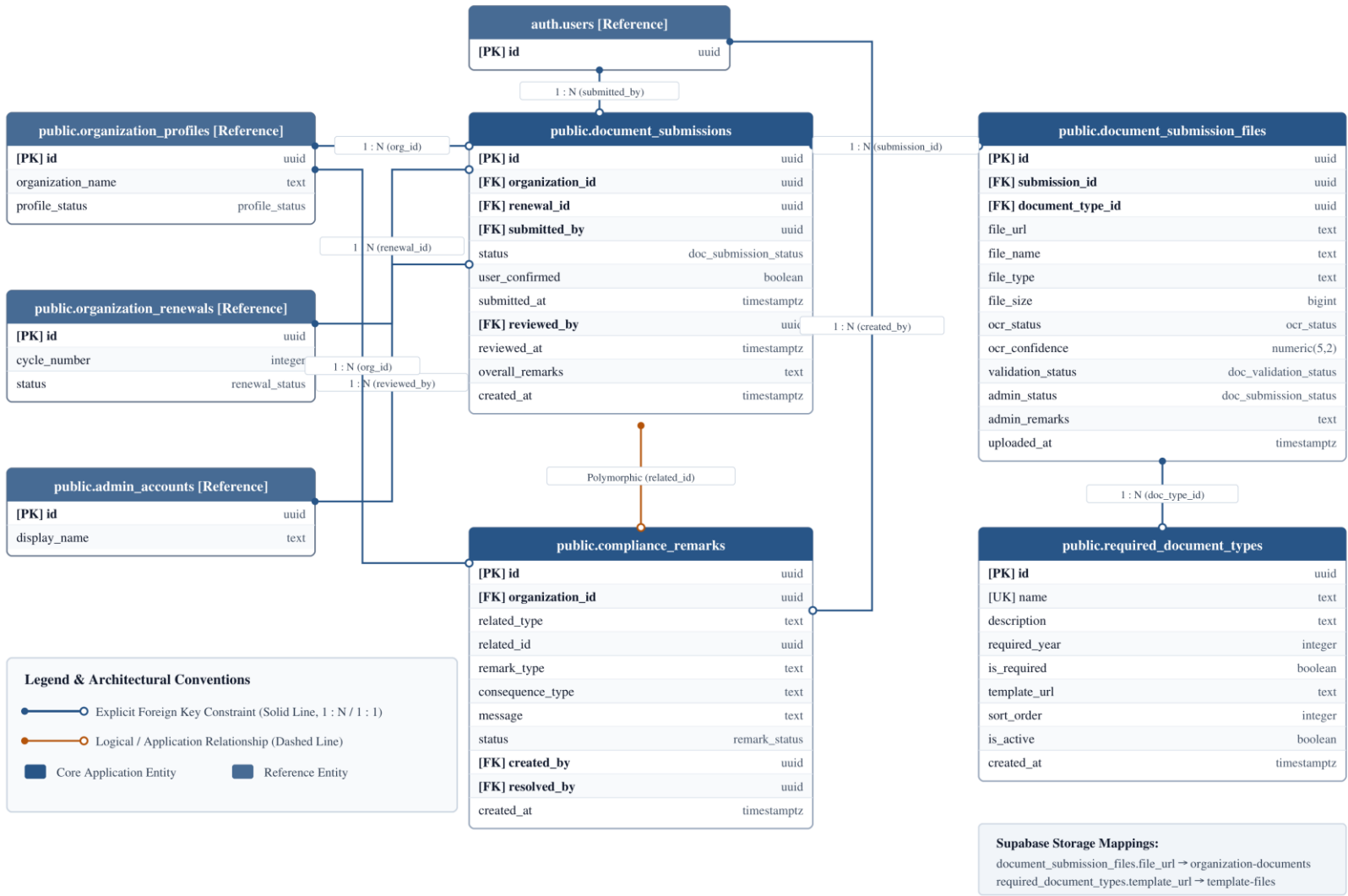
Relationship Catalog for Figure 8.2

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
organization_profiles.user_id	auth.users.id	organization_profiles_user_id_fkey	Explicit FK	One-to-One	CASCADE / NO ACTION
organization_profiles.current_accreditation_id	organization_accreditations.id	organization_profiles_current_accreditation_id_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
organization_accreditations.organization_id	organization_profiles.id	organization_accreditations_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
organization_accreditations.approved_by	admin_accounts.id	organization_accreditations_approved_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
organization_renewals.organization_id	organization_profiles.id	organization_renewals_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
organization_renewals.current_accreditation_id	organization_accreditations.id	organization_renewals_current_accreditation_id_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
organization_renewals.reviewed_by	admin_accounts.id	organization_renewals_reviewed_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
urn_review_history.organization_id	organization_profiles.id	urn_review_history_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
organizations.created_by	auth.users.id	organizations_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION

2.3. Process 2: Manage Compliance Submission and Monitoring

Figure 8.3. Entity Relationship Diagram of Compliance Documents, Requirements, Reviews, and Remarks



Process & Relational Workflow Narrative: Figure 8.3

Figure 8.3 models the documentary compliance, requirement cataloging, administrative verification, and revision workflow. To retain active municipal standing, registered youth organizations must submit mandated annual and periodic documentation. The regulatory requirements are cataloged in `public.required_document_types`, which defines requirement titles (e.g., Accomplishment Report, Financial Statement, Directory of Officers), descriptive instructions, required calendar year, display sort order, and download references to official blank templates hosted in Supabase Storage (`template-files`).

An organization initiates a compliance submission cycle through `public.document_submissions`, which functions as the transactional batch header. This table binds the submission to `organization_profiles(id)`, tracks submission timestamps, user confirmation flags, and overall review status (draft, submitted, under review, approved, needs revision, rejected). If the submission is part of an annual renewal, an explicit foreign key connects to `organization_renewals(id)`. Individual uploaded digital documents are stored in `public.document_submission_files`, establishing foreign keys to both the parent submission and the specific requirement definition. Document records store the Supabase Storage URL (organization-documents bucket), file size, MIME type, optical character recognition (OCR) text extractions, and per-file validation states.

Granular administrative feedback and corrective deficiency notices are captured in `public.compliance_remarks`. This entity links directly to `organization_profiles(id)` and the reviewing officer, while using polymorphic attributes (`related_type` and `related_id`) to target specific submission files or review milestones, maintaining full traceability of resubmission requests.

Table Inventory for Figure 8.3

Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
required_document_types	Master catalogue of mandatory YORP compliance documents and downloadable template references.	id (uuid)	None	template-files	20
document_submissions	Compliance submission batch headers tracking review cycle and administrative decisions.	id (uuid)	organization_id -> organization_profiles.id, submitted_by -> auth.users.id, reviewed_by -> admin_accounts.id, renewal_id -> organization_renewals.id	N/A	6
document_submission_files	Individual uploaded compliance files, storage URIs, OCR metadata, and review states.	id (uuid)	submission_id -> document_submissions.id, document_type_id -> required_document_types.id	organization-documents	39
compliance_remarks	Granular administrative feedback, deficiency notices, and revision instructions.	id (uuid)	organization_id -> organization_profiles.id, created_by -> auth.users.id, resolved_by -> auth.users.id	N/A	0

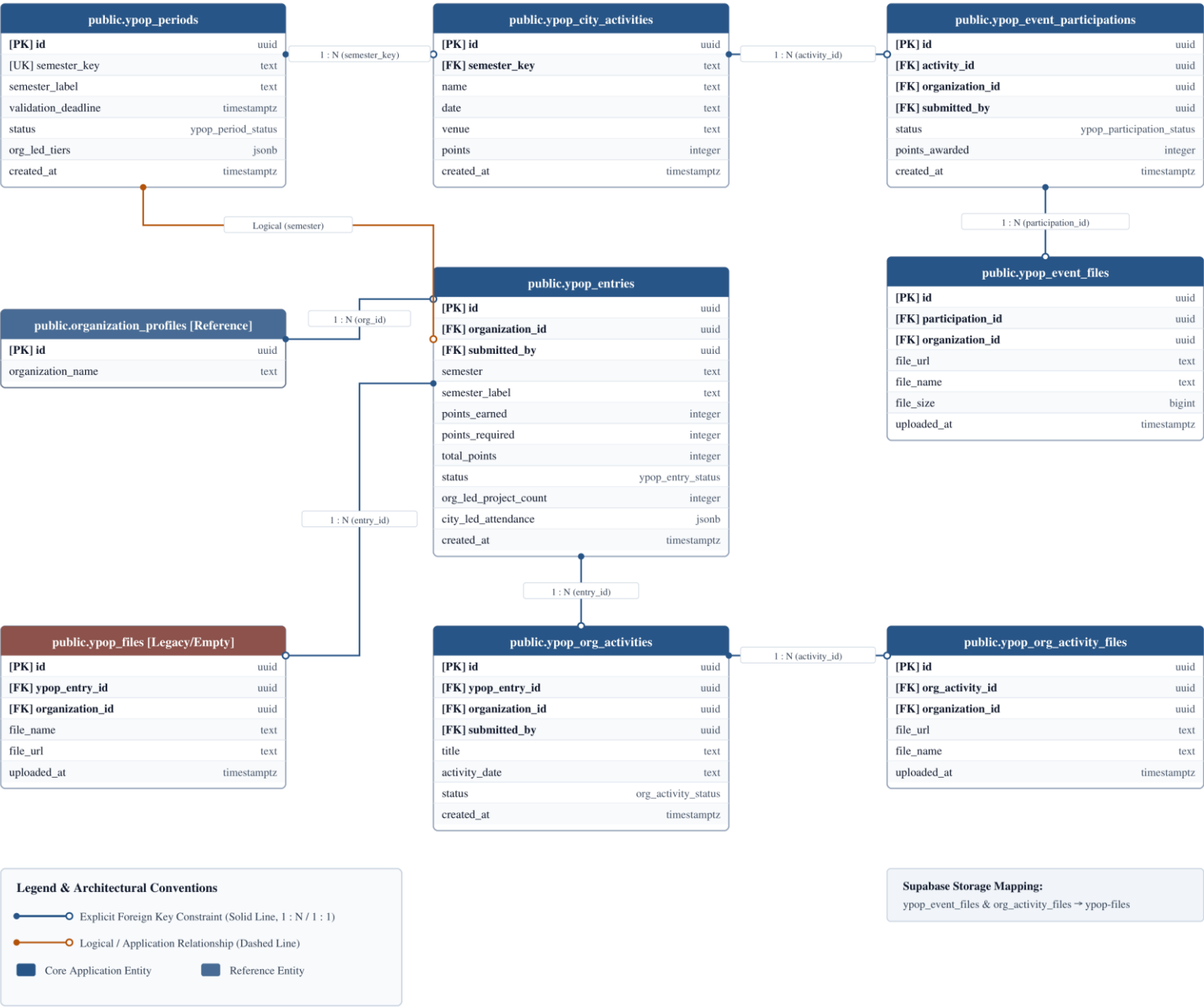
Relationship Catalog for Figure 8.3

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
document_submissions.organization_id	organization_profiles.id	document_submissions_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
document_submissions.submitted_by	auth.users.id	document_submissions_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
document_submissions.reviewed_by	admin_accounts.id	document_submissions_reviewed_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
document_submissions.renewal_id	organization_renewals.id	document_submissions_renewal_id_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
document_submission_files.submission_id	document_submissions.id	document_submission_files_submission_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
document_submission_files.document_type_id	required_document_types.id	document_submission_files_document_type_id_fkey	Explicit FK	Many-to-One	RESTRICT / NO ACTION
compliance_remarks.organization_id	organization_profiles.id	compliance_remarks_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
compliance_remarks.created_by	auth.users.id	compliance_remarks_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
compliance_remarks.resolved_by	auth.users.id	compliance_remarks_resolved_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION

2.4. Process 3: Manage YPOP Participation and Validation

Figure 8.4. Entity Relationship Diagram of YPOP Participation, Activities, and Validation



Process & Relational Workflow Narrative: Figure 8.4

Figure 8.4 depicts the relational data model supporting the Youth Participation Organization Passport (YPOP), an incentive-driven governance mechanism established under Article V, Section 18.1 of the Pasig City Youth Development Code (Ordinance No. 50-2024). YPOP implements a merit-based point accumulation framework that validates youth organization engagement in municipal development programs. The operational framework is anchored by public.ypop_periods, which defines semi-annual passport cycles, validation cutoffs, and JSON-configured organization-led project bonus tiers.

Municipal events organized by PCYDO are cataloged in public.ypop_city_activities, categorized into three distinct point categories: Mandatory City-led activities (4 points), Invitational City-led activities (3 points), and Partnership activities (2 points). Youth organizations log attendance for city activities via public.ypop_event_participations, attaching digital attendance proof, event photos, and certification in public.ypop_event_files (stored in the ypop-files bucket). To incentivize grassroots community work, organizations also log independent community projects in public.ypop_org_activities with supporting verification files in public.ypop_org_activity_files, earning tiered percentage bonuses (10% for 1-3 activities, 15% for 4-6, 20% for 7-9, and 25% for 10 or more activities).

All validated participation points and bonus calculations are consolidated into public.ypop_entries. This master passport entry evaluates whether an organization achieves the mandatory 70% threshold required to qualify for project grant funding. The entity public.ypop_files is maintained as a legacy proof repository linked to ypop_entries for historical data integrity.

Table Inventory for Figure 8.4

Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
ypop_periods	Semi-annual YPOP cycle configuration, validation deadlines, and scoring rules.	id (uuid)	None	N/A	1
ypop_city_activities	City-led municipal activities, scheduled dates, venues, and assignable points.	id (uuid)	semester_key -> ypop_periods.semester_key	N/A	3
ypop_event_participations	Organization attendance logs and participation claims for city-led activities.	id (uuid)	activity_id -> ypop_city_activities.id, organization_id -> organization_profiles.id, submitted_by -> auth.users.id	N/A	6
ypop_event_files	Supporting proof of attendance and event photos for city-led participation.	id (uuid)	participation_id -> ypop_event_participations.id, organization_id -> organization_profiles.id	ypop-files	6
ypop_entries	Master semester passport entries consolidating earned points, bonus tiers, and eligibility status.	id (uuid)	organization_id -> organization_profiles.id, submitted_by -> auth.users.id	N/A	6
ypop_org_activities	Grassroots organization-led PPAs submitted for percentage bonus points.	id (uuid)	ypop_entry_id -> ypop_entries.id, organization_id -> organization_profiles.id, submitted_by -> auth.users.id	N/A	2
ypop_org_activity_files	Proof documents, participant rosters, and post-activity reports for org PPAs.	id (uuid)	org_activity_id -> ypop_org_activities.id, organization_id -> organization_profiles.id	ypop-files	5
ypop_files	Legacy file repository supporting passport entry attachment history.	id (uuid)	ypop_entry_id -> ypop_entries.id, organization_id -> organization_profiles.id	ypop-files	0

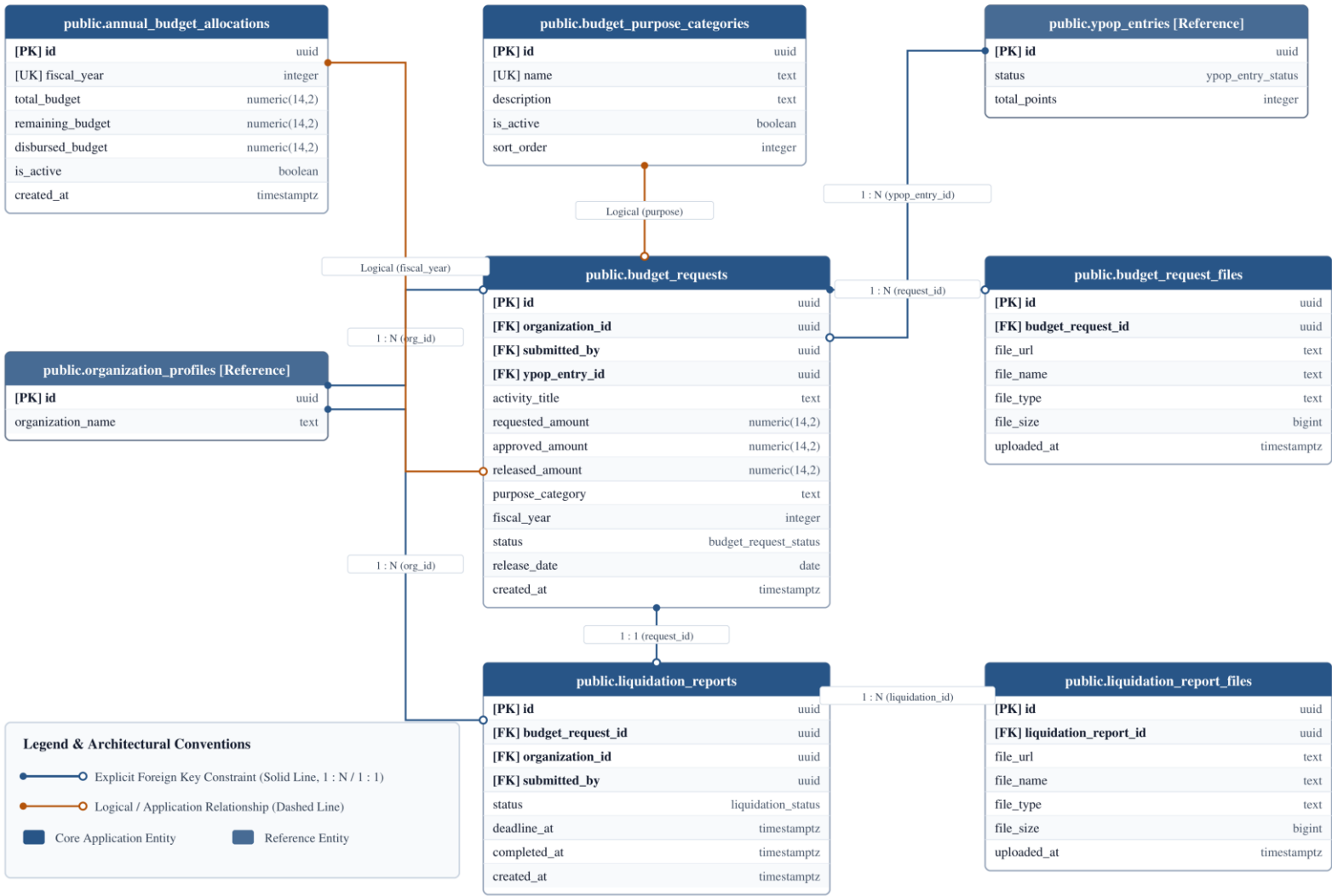
Relationship Catalog for Figure 8.4

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
ypop_city_activities.semester_key	ypop_periods.semester_key	ypop_city_activities_semester_key_fkey	Explicit FK	Many-to-One	CASCADE / CASCADE

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
ypop_event_participations.activity_id	ypop_city_activities.id	ypop_event_participations_activity_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_event_participations.organization_id	organization_profiles.id	ypop_event_participations_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_event_participations.submitted_by	auth.users.id	ypop_event_participations_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
ypop_event_files.participation_id	ypop_event_participations.id	ypop_event_files_participation_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_event_files.organization_id	organization_profiles.id	ypop_event_files_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_entries.organization_id	organization_profiles.id	ypop_entries_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_entries.submitted_by	auth.users.id	ypop_entries_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
ypop_org_activities.ypop_entry_id	ypop_entries.id	ypop_org_activities_ypop_entry_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_org_activities.organization_id	organization_profiles.id	ypop_org_activities_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_org_activities.submitted_by	auth.users.id	ypop_org_activities_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
ypop_org_activity_files.org_activity_id	ypop_org_activities.id	ypop_org_activity_files_org_activity_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_org_activity_files.organization_id	organization_profiles.id	ypop_org_activity_files_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_files.ypop_entry_id	ypop_entries.id	ypop_files_ypop_entry_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
ypop_files.organization_id	organization_profiles.id	ypop_files_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

2.5. Process 4 & 5: Manage Budget Requests, Approval, and Liquidation Reporting

Figure 8.5. Entity Relationship Diagram of Budget Requests, Budget Allocation, Release, and Liquidation



Process & Relational Workflow Narrative: Figure 8.5

Figure 8.5 illustrates the financial governance subsystem managing municipal youth project grants, appropriation tracking, fund disbursement, and statutory liquidation oversight. Under Ordinance No. 50-2024, YPOP-qualified organizations may apply for project grants of up to Php 100,000.00. The fiscal ceiling is administered through public.annual_budget_allocations, which records annual budget envelopes, allocated sums, released funds, and cumulative liquidation totals. Standardized expenditure classifications are maintained in public.budget_purpose_categories (e.g., capability building, sports and health development, community outreach).

An eligible organization submits a formal funding request via public.budget_requests. This table validates qualifying status via an explicit foreign key to public.ypop_entries(id) and public.organization_profiles(id). Requests capture proposed activity dates, venue, requested amount, approved amount, release date, and administrative review states (pending, approved, onsite_submission, released, rejected). Supporting project proposals and itemized cost matrices are uploaded to public.budget_request_files and persisted in the budget-request-files storage bucket.

Upon administrative fund release, a strictly bound 1:1 liquidation report is instantiated in public.liquidation_reports. In accordance with City Accounting Office (CAO) and Commission on Audit (COA) circulars, recipient organizations must submit comprehensive financial liquidations within ten (10) working days post-implementation. Organizations upload scanned official receipts, honoraria payrolls, and disbursement vouchers to public.liquidation_report_files (stored in liquidation-report-files). Administrators track compliance deadlines, evaluate submitted evidence, and record final liquidation approval, ensuring complete fiscal transparency.

Table Inventory for Figure 8.5

Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
annual_budget_allocations	Fiscal year municipal appropriation envelopes, allocated totals, and disbursement balances.	id (uuid)	None	N/A	1
budget_purpose_categories	Standardized expenditure categories governing allowable project design costs.	id (uuid)	None	N/A	8
budget_requests	Formal grant funding applications, approved amounts, review states, and fund release logs.	id (uuid)	organization_id -> organization_profiles.id, submitted_by -> auth.users.id, ypop_entry_id -> ypop_entries.id	N/A	5
budget_request_files	Uploaded project design briefs, cost itemizations, and endorsement files.	id (uuid)	budget_request_id -> budget_requests.id	budget-request-files	5
liquidation_reports	Statutory post-activity financial liquidation reports linked 1:1 to approved budget requests.	id (uuid)	budget_request_id -> budget_requests.id, organization_id -> organization_profiles.id, submitted_by -> auth.users.id	N/A	3
liquidation_report_files	Scanned official receipts, disbursement vouchers, and honoraria payrolls.	id (uuid)	liquidation_report_id -> liquidation_reports.id	liquidation-report-files	3

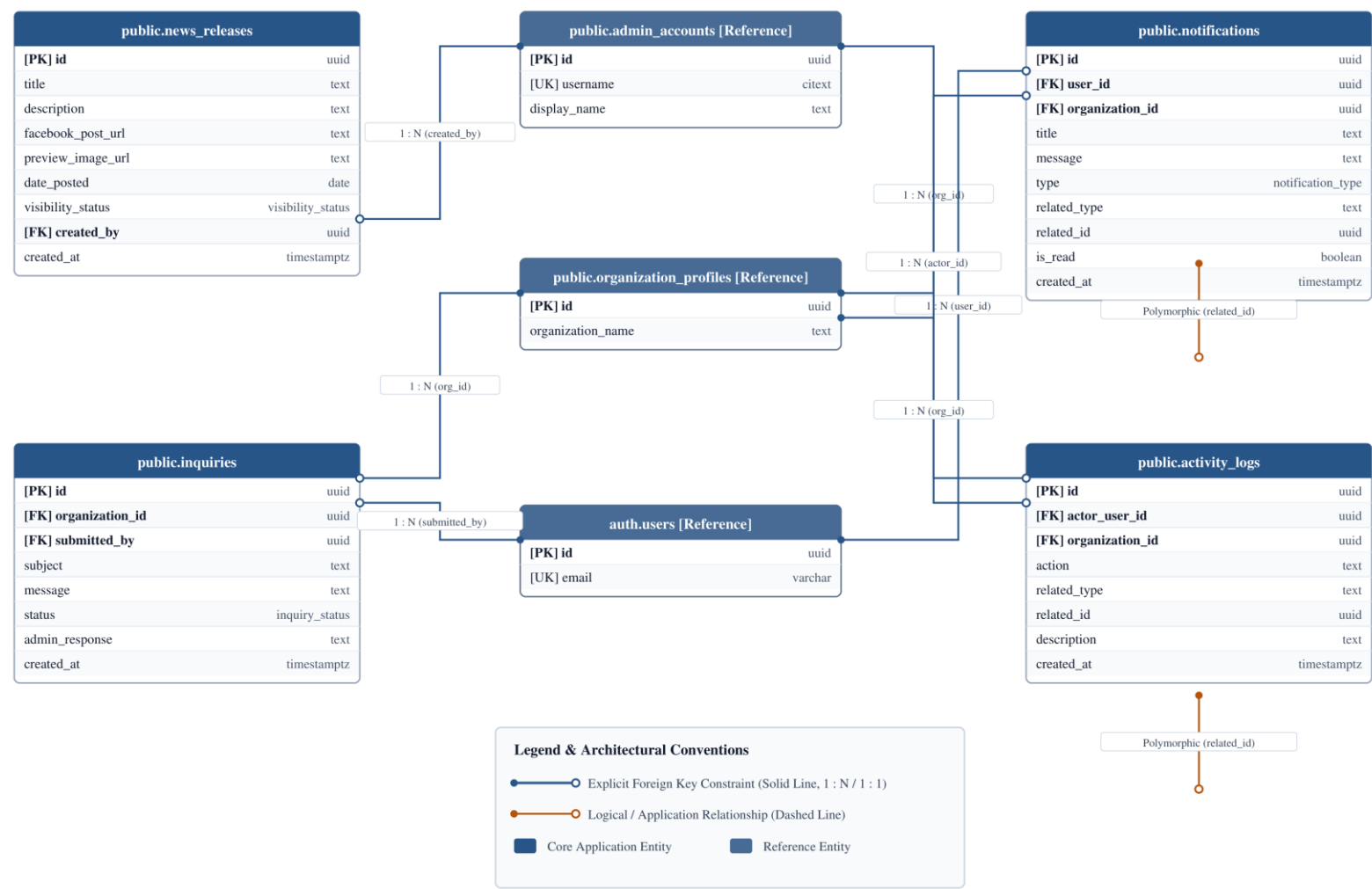
Relationship Catalog for Figure 8.5

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
budget_requests.organization_id	organization_profiles.id	budget_requests_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
budget_requests.submitted_by	auth.users.id	budget_requests_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
budget_requests.ypop_entry_id	ypop_entries.id	budget_requests_ypop_entry_id_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
budget_request_files.budget_request_id	budget_requests.id	budget_request_files_budget_request_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
liquidation_reports.budget_request_id	budget_requests.id	liquidation_reports_budget_request_id_fkey	Explicit FK	One-to-One	CASCADE / NO ACTION
liquidation_reports.organization_id	organization_profiles.id	liquidation_reports_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
liquidation_reports.submitted_by	auth.users.id	liquidation_reports_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
liquidation_report_files.liquidation_report_id	liquidation_reports.id	liquidation_report_files_liquidation_report_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

2.6. Process 6 & 7: Communications, System Notifications, and Administrative Audit Trail

Figure 8.6. Entity Relationship Diagram of News, Notifications, Inquiries, and Activity Logs



Process & Relational Workflow Narrative: Figure 8.6

Figure 8.6 represents the cross-cutting communication, event notification, user support inquiry, and security audit subsystem. Public communication and portal announcements are managed via public.news_releases, authored by administrative personnel (admin_accounts.id) with publication flags, posting dates, and media image paths in Supabase Storage (news-release-images bucket).

Workflow responsiveness and regulatory deadline awareness are maintained through public.notifications. This entity delivers in-app alerts to organization representatives (auth.users.id) and tracks unread states. To dynamically reference diverse operational triggers without creating multiple hard foreign key constraints, notifications employs polymorphic association attributes: related_type (e.g., 'budget_requests', 'liquidation_reports', 'document_submissions') and related_id (referencing the UUID of the target record).

Technical support and administrative inquiries are handled via public.inquiries, allowing youth leaders to submit questions, track status, and receive structured responses from PCYDO personnel. Administrative accountability and security auditability are guaranteed by public.activity_logs. This table records an append-only audit trail of critical operational actions (logins, status changes, approvals, financial modifications), capturing the acting administrator (admin_accounts.id), the impacted organization (organization_profiles.id), polymorphic workflow entity targets, descriptive text, and precise timestamping.

Table Inventory for Figure 8.6

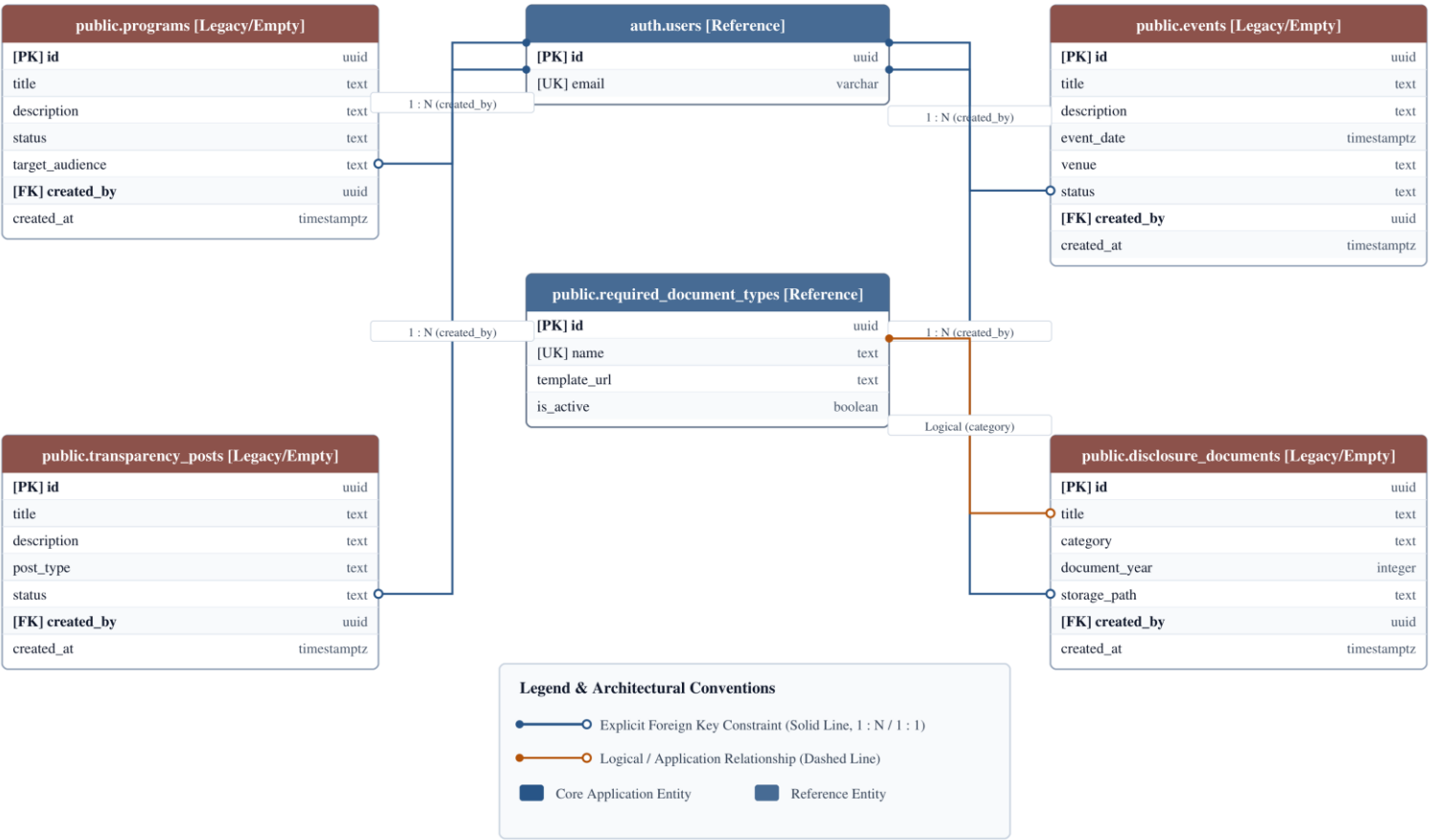
Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
news_releases	Official municipal news bulletins, youth program announcements, and media banners.	id (uuid)	created_by -> admin_accounts.id	news-release-images	3
notifications	Automated system notifications and deadline alerts delivered to authenticated users.	id (uuid)	organization_id -> organization_profiles.id, user_id -> auth.users.id	N/A	49
inquiries	Technical support tickets and administrative inquiry threads submitted by organizations.	id (uuid)	organization_id -> organization_profiles.id, submitted_by -> auth.users.id	N/A	8
activity_logs	Tamper-evident administrative audit trail capturing user actions, timestamps, and operations.	id (uuid)	actor_user_id -> admin_accounts.id, organization_id -> organization_profiles.id	N/A	373

Relationship Catalog for Figure 8.6

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
news_releases.created_by	admin_accounts.id	news_releases_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
notifications.organization_id	organization_profiles.id	notifications_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
notifications.user_id	auth.users.id	notifications_user_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
inquiries.organization_id	organization_profiles.id	inquiries_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
inquiries.submitted_by	auth.users.id	inquiries_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
activity_logs.actor_user_id	admin_accounts.id	activity_logs_actor_user_id_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
activity_logs.organization_id	organization_profiles.id	activity_logs_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

2.7. Process 6: Manage Forms, Templates, News, and Inquiries (Sub-process: Public Engagement & Transparency)

Figure 8.7. Entity Relationship Diagram of Forms, Templates, Programs, Events, and Transparency Records



Process & Relational Workflow Narrative: Figure 8.7

Figure 8.7 delineates the public portal, civic engagement, transparency disclosure, and community event cataloging entities. To promote good governance and open data in local government administration, Y-TRACE provides public repositories for municipal transparency records and youth development programs.

Public educational and youth welfare programs are cataloged in public.programs, tracking program descriptions, target demographics, and administering personnel. Municipal youth gatherings and civic summits are scheduled in public.events, recording event dates, venues, and registration links. Civic transparency and statutory municipal disclosures are published via public.transparency_posts and public.disclosure_documents, providing public citizens and youth stakeholders with downloadable access to official municipal reports, annual budgets, and resolutions stored in the transparency-attachments storage bucket. All four entities maintain explicit foreign keys to auth.users(id) to record the publishing officer. Downloadable document templates are cross-referenced from public.required_document_types, which hosts official blank compliance forms in the template-files bucket.

Table Inventory for Figure 8.7

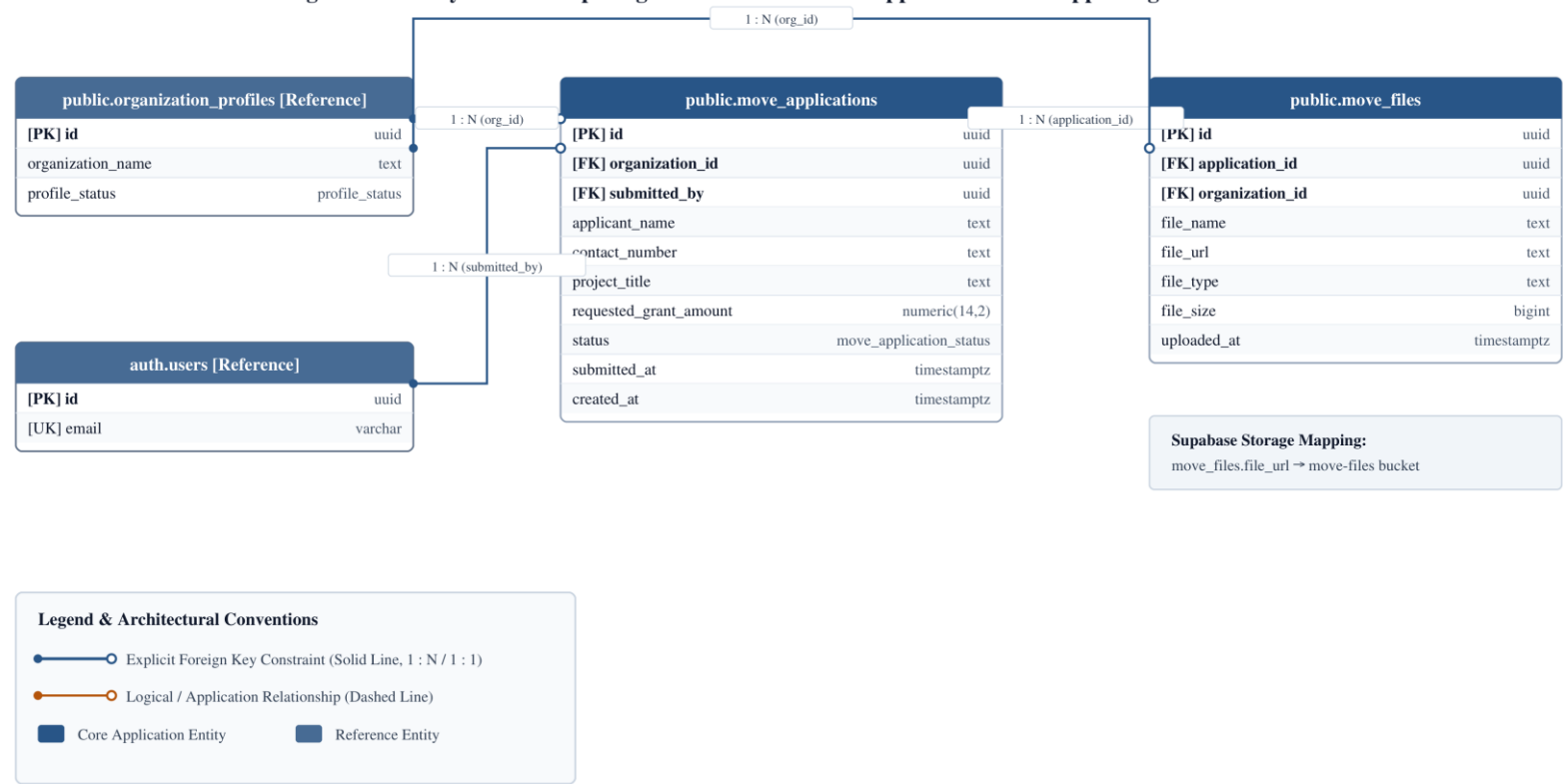
Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
programs	Institutional youth programs, target youth demographics, and program outlines.	id (uuid)	created_by -> auth.users.id	N/A	0
events	Municipal youth events calendar, dates, venues, and registration URLs.	id (uuid)	created_by -> auth.users.id	N/A	0
transparency_posts	Public governance posts and administrative transparency announcements.	id (uuid)	created_by -> auth.users.id	N/A	0
disclosure_documents	Public downloadable disclosure PDFs, resolutions, and budget summaries.	id (uuid)	created_by -> auth.users.id	transparency-attachments	0

Relationship Catalog for Figure 8.7

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
programs.created_by	auth.users.id	programs_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
events.created_by	auth.users.id	events_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
transparency_posts.created_by	auth.users.id	transparency_posts_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
disclosure_documents.created_by	auth.users.id	disclosure_documents_created_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION

2.8. Process 1 & 6: Specialized Youth MOVE Assistance Program

Figure 8.8. Entity Relationship Diagram of Youth MOVE Applications and Supporting Documents



Process & Relational Workflow Narrative: Figure 8.8

Figure 8.8 depicts the dedicated database schema partition supporting the Youth MOVE (Mobility Opportunity for Vibrant Engagement) program. Youth MOVE is a specialized municipal initiative providing youth organizations with vehicular logistics, travel coordination, and mobility support for educational, athletic, and leadership representation outside Pasig City.

An accredited organization submits a mobility request through public.move_applications, creating explicit foreign keys to public.organization_profiles(id) and public.auth.users(id). The application records travel dates, destinations, passenger counts, vehicle requirements, and administrative review states (pending, approved, dispatched, completed, rejected). Supporting documentation — such as parental consent waivers, official competition invitations, and passenger manifests — are tracked in public.move_files, with physical files stored securely in the move-files storage bucket. Explicit foreign keys enforce referential integrity between move_files, the parent application, and the participating organization profile.

Table Inventory for Figure 8.8

Table Name	Functional Role / Purpose	Primary Key	Foreign Keys	Storage Bucket	Audited Rows
move_applications	Specialized youth mobility and vehicle coordination applications.	id (uuid)	organization_id -> organization_profiles.id, submitted_by -> auth.users.id	N/A	0
move_files	Supporting travel waivers, passenger rosters, and event invitations.	id (uuid)	application_id -> move_applications.id, organization_id -> organization_profiles.id	move-files	0

Relationship Catalog for Figure 8.8

Child Table & Column	Parent Table & Column	Constraint Name	Type	Cardinality	On Delete / Update
move_applications.organization_id	organization_profiles.id	move_applications_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
move_applications.submitted_by	auth.users.id	move_applications_submitted_by_fkey	Explicit FK	Many-to-One	SET NULL / NO ACTION
move_files.application_id	move_applications.id	move_files_application_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION
move_files.organization_id	organization_profiles.id	move_files_organization_id_fkey	Explicit FK	Many-to-One	CASCADE / NO ACTION

3. Cross-Process and Polymorphic Relationship Architecture

While the physical relational integrity of the Y-TRACE database is strictly enforced through sixty-five (65) foreign key constraints, several vital enterprise workflows require cross-cutting associations that cannot be modeled through conventional single-target foreign keys. To preserve structural flexibility, avoid proliferation of sparse nullable foreign key columns, and prevent tight coupling between independent modules, the architecture incorporates two specialized relational design patterns: Polymorphic Application References and Logical / Non-Enforced References.

3.1. Polymorphic Application References

In multi-functional local governance systems, notification mechanisms, administrative audit trails, and review remarks must dynamically attach to heterogeneous records across different operational modules. Rather than creating disparate notification tables for every individual entity or polluting tables with dozens of nullable foreign keys, Y-TRACE implements the Polymorphic Association pattern. Under this pattern, a record stores two complementary attributes: a discriminator column (related_type as text) that designates the target entity name, and an identifier column (related_id as uuid) that stores the target record's primary key value. Application logic and UI routing resolve these pairs to direct users to the exact submission, report, or inquiry in question.

Source Table	Discriminator Columns	Supported Target Entities	Operational Workflow Purpose
notifications	related_type (text) related_id (uuid)	budget_requests, liquidation_reports, document_submissions, ypop_entries, inquiries	Directs users to specific workflow milestones, revision requests, or approval notices across all modules.
activity_logs	related_type (text) related_id (uuid)	organization_profiles, budget_requests, liquidation_reports, document_submissions, ypop_entries, policy_versions	Provides immutable audit trails capturing administrative actions (approvals, edits, logins) against target entities.
compliance_remarks	related_type (text) related_id (uuid)	document_submissions, document_submission_files, organization_renewals	Attaches granular corrective feedback and deficiency notices to specific compliance submissions or files.

Table 8.9. Polymorphic Relationship Specifications in Y-TRACE

3.2. Logical / Application-Level Cross-Table References

Certain business relationships in Y-TRACE are purposely maintained as logical links at the application layer rather than hard PostgreSQL foreign keys. This architectural decision accommodates municipal data lifecycles where historical records must remain immutable and decoupled from active configuration changes, such as annual budget resets or category taxonomy refinements. Table 8.10 documents the verified logical relationships.

Source Field	Target Field	Relationship Nature	Architectural Rationale
budget_requests.fiscal_year	annual_budget_allocations.fiscal_year	Logical Integer Link	Binds grant requests to annual appropriations without locking prior fiscal allocations during annual rollover.
budget_requests.purpose_category	budget_purpose_categories.name	Logical Text Key Link	Preserves verbatim proposal category strings even if administrative taxonomy definitions are reworded.
ypop_entries.semester	ypop_periods.semester_key	Logical Text Key Link	Maintains historical student passport records linked to semester codes (e.g. '2026-1st-sem') without hard cascade dependency.
organization_profiles.barangay	barangays.name	Logical Name Reference	Facilitates geographic filtering across Pasig City's 30 barangays while permitting custom settlement names if required.
user_profiles.barangay	barangays.name	Logical Name Reference	Associates user residential location with official barangay master list for demographic analytics.

Table 8.10. Logical and Non-Enforced Application References

3.3. Shared Core Entities Across Process Workflows

A critical strength of the Y-TRACE schema architecture is the centralization of core anchor entities across multiple workflows. Rather than duplicating organizational or identity metadata, the schema utilizes shared master entities that act as parents across diverse processes:

1. `public.organization_profiles`: Anchors eight (8) separate subsystems: Registration (Fig 8.2), Compliance (Fig 8.3), YPOP (Fig 8.4), Budget (Fig 8.5), Liquidation (Fig 8.5), Notifications & Audit (Fig 8.6), and Youth MOVE (Fig 8.8).
2. `auth.users`: Provides universal actor identity and authentication across all public portal entities (Fig 8.1 through 8.8).
3. `public.admin_accounts`: Serves as the universal administrative actor reference across review, approval, resolution, and audit logging modules.

4. Supabase Storage Bucket Architecture & Object Integration

To handle digital documentary submissions, financial evidence, scanned official receipts, and media attachments without bloat in the PostgreSQL relational engine, Y-TRACE integrates Supabase Storage. The system provisions nine (9) dedicated storage buckets, each partitioned with specific access-control policies, MIME restrictions, and automated path generation. Storage URIs are stored in corresponding database columns to maintain 1:1 or 1:N relational links.

Bucket Name	Access Type	Max File Size	Linked Table & Column	Workflow & Path Structure
organization-documents	Private (Auth)	15 MB	document_submission_files.file_url	Mandatory annual compliance files (Accomplishment Reports, Roster). Path: /{org_id}/{submission_id}/{filename}
budget-request-files	Private (Auth)	20 MB	budget_request_files.file_url	Project grant proposals, activity designs, itemized budgets. Path: /{org_id}/{request_id}/{filename}
liquidation-report-files	Private (Auth)	25 MB	liquidation_report_files.file_url	Official receipts, disbursement vouchers, honoraria payrolls. Path: /{org_id}/{liquidation_id}/{filename}
ypop-files	Private (Auth)	15 MB	ypop_event_files.file_url ypop_org_activity_files.file_url	Proof of attendance, event photos, activity reports for passport scoring. Path: /{org_id}/ypop/{entry_id}/{filename}
move-files	Private (Auth)	15 MB	move_files.file_url	Youth MOVE vehicle applications, passenger rosters, liability waivers. Path: /{org_id}/move/{application_id}/{filename}
template-files	Public (Read)	10 MB	required_document_types.template_url	Downloadable blank compliance forms, liquidation matrices, and guides. Path: /templates/{filename}
news-release-images	Public (Read)	5 MB	news_releases.preview_image_url	Promotional banners, cover images, and media assets for portal news. Path: /news/{news_id}/{filename}
transparency-attachments	Public (Read)	25 MB	disclosure_documents.storage_path	Official municipal youth codes, budget allocations, audit resolutions. Path: /transparency/{year}/{filename}
avatars	Public (Read)	2 MB	user_profiles.avatar_url	User profile photos and organization logos. Path: /avatars/{user_id}/{filename}

Table 8.11. Supabase Storage Bucket Mapping and Configuration Specifications

5. Database Schema Coverage Matrix (42 of 42 Application Tables)

To satisfy the strict empirical requirements of the capstone technical evaluation, Table 8.12 presents the complete Database Schema Coverage Matrix. Every single one of the forty-two (42) public application tables identified during the authoritative Supabase database audit (conducted September 15, 2026) is cataloged alongside its primary operational process, assigned ERD figure, physical architectural role, audited live row count, and verified inclusion status. Zero tables are omitted, ensuring 100.0% comprehensive documentation.

#	Table Name	Primary Operational Process	Figure	Entity Architectural Role	Row Count	Included?
1	activity_logs	Process 7: Manage Administration & Reporting	Figure 8.6	Core Administrative Audit Trail	373	YES
2	admin_accounts	Process 7: Manage Administration & Reporting	Figure 8.1	Core Administrative Credentials & Profile	5	YES
3	admin_sessions	Process 7: Manage Administration & Reporting	Figure 8.1	Core Server-Side Session Tracking	241	YES
4	annual_budget_allocations	Process 4 & 5: Budget Requests & Liquidation	Figure 8.5	Core Fiscal Ceiling & Appropriation	1	YES
5	barangays	Process 1: Organization Registration & Renewal	Figure 8.2	Core Geographic & Demographic Reference	3	YES
6	budget_purpose_categories	Process 4: Budget Requests & Approval	Figure 8.5	Core Allowable Cost Classification	8	YES
7	budget_request_files	Process 4: Budget Requests & Approval	Figure 8.5	Core Proposal Document Attachments	5	YES
8	budget_requests	Process 4: Budget Requests & Approval	Figure 8.5	Core Grant Proposal & Release Workflow	5	YES
9	compliance_remarks	Process 2: Compliance Submission & Monitoring	Figure 8.3	Workflow Deficiency & Revision Tracking	0	YES
10	disclosure_documents	Process 6: Forms, Templates, News, Transparency	Figure 8.7	Public Transparency Document Store	0	YES
11	document_submission_files	Process 2: Compliance Submission & Monitoring	Figure 8.3	Core Uploaded Compliance Attachments	39	YES
12	document_submissions	Process 2: Compliance Submission & Monitoring	Figure 8.3	Core Compliance Submission Batch Header	6	YES
13	events	Process 6: Forms, Templates, News, Transparency	Figure 8.7	Public Municipal Youth Calendar	0	YES
14	inquiries	Process 6 & 7: Communications & Support	Figure 8.6	Two-Way User Inquiry Threading	8	YES
15	liquidation_report_files	Process 5: Liquidation Reporting & Utilization	Figure 8.5	Core Scanned Official Receipt Store	3	YES
16	liquidation_reports	Process 5: Liquidation Reporting & Utilization	Figure 8.5	Core CAO/COA Post-Activity Liquidation	3	YES
17	move_applications	Process 1 & 6: Specialized Youth MOVE	Figure 8.8	Youth Mobility & Logistics Application	0	YES
18	move_files	Process 1 & 6: Specialized Youth MOVE	Figure 8.8	Mobility Roster & Waiver Attachments	0	YES
19	news_releases	Process 6: Forms, Templates, News, Transparency	Figure 8.6	Official Municipal News Bulletin	3	YES
20	notifications	Process 6 & 7: Communications & Alerts	Figure 8.6	Cross-Cutting In-App Workflow Alerts	49	YES
21	organization_accreditations	Process 1: Organization Registration & Renewal	Figure 8.2	Official Municipal Accreditation Certificate	5	YES
22	organization_profiles	Process 1: Organization Registration & Renewal	Figure 8.2	Primary Youth Organization Master Registry	6	YES
23	organization_renewals	Process 1: Organization Registration & Renewal	Figure 8.2	Annual YORP Renewal Application Cycle	0	YES
24	organizations	Process 1: Organization Registration & Renewal	Figure 8.2	Legacy Organization Fallback Store	0	YES
25	policy_versions	Process 7: Manage Administration & Reporting	Figure 8.1	Legal Terms & Privacy Policy Versions	3	YES
26	programs	Process 6: Forms, Templates, News, Transparency	Figure 8.7	Municipal Youth Development Programs	0	YES
27	required_document_types	Process 2: Compliance Submission & Monitoring	Figure 8.3	Mandatory Compliance Document Catalog	20	YES
28	roles	Process 7: Manage Administration & Reporting	Figure 8.1	Role-Based Access Control (RBAC) Master	4	YES
29	transparency_posts	Process 6: Forms, Templates, News, Transparency	Figure 8.7	Public Civic Governance Bulletins	0	YES
30	units	Process 7: Manage Administration & Reporting	Figure 8.1	PCYDO Functional Administrative Units	3	YES

#	Table Name	Primary Operational Process	Figure	Entity Architectural Role	Row Count	Included?
31	urn_review_history	Process 1: Organization Registration & Renewal	Figure 8.2	Unique Registration Number Audit Trail	0	YES
32	user_policy_acceptance	Process 7: Manage Administration & Reporting	Figure 8.1	RA 10173 Legal Acceptance Gate Log	9	YES
33	user_profiles	Process 7: Manage Administration & Reporting	Figure 8.1	Biographical Profile for Authenticated Users	20	YES
34	user_roles	Process 7: Manage Administration & Reporting	Figure 8.1	Associative User-to-Role Assignment	20	YES
35	ypop_city_activities	Process 3: YPOP Participation & Validation	Figure 8.4	City-Led Activities & Point Values	3	YES
36	ypop_entries	Process 3: YPOP Participation & Validation	Figure 8.4	Master Semester YPOP Passport Header	6	YES
37	ypop_event_files	Process 3: YPOP Participation & Validation	Figure 8.4	Attendance Verification File Store	6	YES
38	ypop_event_participations	Process 3: YPOP Participation & Validation	Figure 8.4	Organization Attendance Claim Headers	6	YES
39	ypop_files	Process 3: YPOP Participation & Validation	Figure 8.4	Legacy Passport Proof File Store	0	YES
40	ypop_org_activities	Process 3: YPOP Participation & Validation	Figure 8.4	Grassroots Community PPA Records	2	YES
41	ypop_org_activity_files	Process 3: YPOP Participation & Validation	Figure 8.4	Grassroots Project Evidence & Photos	5	YES
42	ypop_periods	Process 3: YPOP Participation & Validation	Figure 8.4	YPOP Semi-Annual Cycle Configuration	1	YES

Table 8.12. Complete Database Schema Coverage Matrix

FORMAL AUDIT VERIFICATION DECLARATION: 42 / 42 APPLICATION TABLES DOCUMENTED

All forty-two (42) public application tables and sixty-five (65) foreign key constraints identified in the authoritative Supabase audit have been rigorously mapped into process-oriented Entity Relationship Diagrams (Figures 8.1 through 8.8). Zero application tables were omitted. All standby, reference, and legacy entities are fully integrated and annotated in accordance with capstone research standards.

6. Database Integrity, Security & Governance Architecture

The Y-TRACE database enforces a multi-layered security and data integrity perimeter combining declarative SQL constraints, Row-Level Security (RLS) policies, database triggers, and PostgreSQL stored routines.

6.1. Row-Level Security (RLS) Enforcement

Every application table in the public schema has Row-Level Security actively enabled (`ALTER TABLE ... ENABLE ROW LEVEL SECURITY`). Fifty-eight (58) granular RLS policies govern SELECT, INSERT, UPDATE, and DELETE privileges based on JWT claims and administrative roles:

- **Organization Multi-Tenant Isolation:** Youth organizations are restricted to viewing and mutating rows where `organization_id` matches their authenticated profile ID or where `submitted_by` equals `auth.uid()`.
- **Administrative Supervisory Access:** Authenticated staff accounts belonging to PCYDO administrative units possess elevated access privileges verified through role checking functions (`is_admin()`, `is_super_admin()`), allowing cross-organizational review, approval, and audit actions.
- **Public Read Policies:** Tables supporting the public portal (`news_releases`, `required_document_types`, `transparency_posts`, `programs`, `events`) grant unrestricted SELECT access to anonymous and authenticated users when active flags are set to true.

6.2. Database Triggers and Automation

Thirty (30) PostgreSQL database triggers automate crucial audit and lifecycle behaviors:

- **Timestamp Synchronization:** Triggers execute `set_updated_at()` BEFORE UPDATE on tables to maintain tamper-evident audit timestamps.
- **Administrative Action Logging:** Triggers on `budget_requests`, `document_submissions`, and `ypop_entries` automatically write immutable entries to `public.activity_logs` whenever an approval status transition occurs.
- **Notification Dispatch:** Status changes in grant requests or compliance evaluations fire asynchronous notification triggers, inserting targeted alerts into `public.notifications` for immediate user feedback.

6.3. Stored Functions and Stored Procedures (RPCs)

The database houses 120 custom stored functions and RPC endpoints executing business-critical calculations inside the PostgreSQL engine:

- `calculate_ypop_score(entry_id uuid)`: Aggregates attended city-led activity points, validates attendance proof, adds org-led PPA bonus percentage tiers, and determines whether the 70% threshold is satisfied.
- `get_my_organization_profile()`: Secure RPC returning complete organizational metadata for the currently authenticated user session.
- `sync_admin_user()`: Coordinates credentials between `auth.users` and `public.admin_accounts` for unified administrative lifecycle management.